5D is the Case where the curled geometry is a circle, S^1.

Notation and Setup:

Starting Metric for this Case: g_{00}=A[t, r], g_{11}=-a[t]^2 B[r], g_{22}=-r^2 a[t]^2, g_{33}=-r^2 a[t]^2 Sin[θ]^2, g_{44}=-a[t]^2 p_4^2 $\mathcal{D}_4$[r], a(t) not constant

Corresponding Line Element

ds^2 = A[t, r] dt^2 - a[t]^2 B[r] dr^2 - r^2 a[t]^2 $d\theta^2$ - r^2 a[t]^2 Sin[θ]^2 $d\phi^2$ - a[t]^2 p_4^2 $\mathcal{D}_4$[r] $d\psi_4^2$

Output Display Simplification - Parameters are sometimes suppressed
Prime (') means derivative wrt r.
Dot (.) means derivative wrt t.
Examples: **A = A(t,r)**, **A'=∂_rA(t,r)** and **$\dot{A}$=∂_tA(t,r)**

Lambda vacuum equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameter a, (2) C = 1, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}$(t) = c^2 is the Friedmann Continuity Equation (which generates the a(t) formula).

FG-GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and C, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}$(t) = c^2 is the Friedmann Continuity Equation (which generates the a(t) formula).

a: GRW Equations

The line element above shows the initial assumptions we have made:

a(t) is a free parameter,

A(t,r) is a function of t and r,

B(r) is strictly a function of r,

C(r) = 1 (required by definition, above),

and $\mathcal{D}_4$(r) is assumed to be strictly a function of r.

Notes:

• Unlike 4D, FG-GRW is too difficult to solve for directly, so we first

solve for GRW.

- We do not assume formulas for A, B, or $\mathbb{D}_4$ but rather solve for them using the field equations.
- We 'allow' but don't require the curled space metric $\mathbb{D}_4$ to be sensitive to r.

With only parameter C initially set, the unknowns are A(t,r), B(r), $\mathbb{D}_4$(r), and the scale factor a(t).

Using this starting point, the non-zero Lambda field equations become

$$R_{00}: \quad \frac{4\,\ddot{a}}{a\,A} - \frac{2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{a^2\,A\,r\,B} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{A'\,B'}{4\,a^2\,A\,B^2} - \frac{A'\,\mathbb{D}_4'}{4\,a^2\,A\,B\,\mathbb{D}_4} - \frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{3\,\dot{a}\,A'}{2\,a\,A} = 0$$

$$R_{11}: \quad \frac{3\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{A'\,B'}{4\,a^2\,A\,B^2} + \frac{B'\,\mathbb{D}_4'}{4\,a^2\,B^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,B\,\mathbb{D}_4^2} - \frac{A''}{2\,a^2\,A\,B} - \frac{\mathbb{D}_4''}{2\,a^2\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{3\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathbb{D}_4'}{2\,a^2\,r\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{33}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{3\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathbb{D}_4'}{2\,a^2\,r\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}_4'}{a^2\,r\,B\,\mathbb{D}_4} - \frac{A'\,\mathbb{D}_4'}{4\,a^2\,A\,B\,\mathbb{D}_4} + \frac{B'\,\mathbb{D}_4'}{4\,a^2\,B^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,B\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{2\,a^2\,B\,\mathbb{D}_4} = \Lambda$$

R_{33} is redundant, the same as R_{22}.

Since a(t) ≠ constant, from R_{01} we get that

A(t,r) = $\mathcal{A}$(t), a new unknown, strictly a function of t.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{4\,\ddot{a}}{a\,\mathcal{A}} - \frac{2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{3\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,\mathbb{D}_4'}{4\,a^2\,B^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,B\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{2\,a^2\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{3\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathbb{D}_4'}{2\,a^2\,r\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{\mathbb{D}_4'}{a^2\,r\,B\,\mathbb{D}_4} + \frac{B'\,\mathbb{D}_4'}{4\,a^2\,B^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,B\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{2\,a^2\,B\,\mathbb{D}_4} = \Lambda$$

We can immediately find $\mathcal{A}$ from R_{00}:

$$\mathcal{H}(t) = \frac{4\,\dot{a}^2}{a^2\,\Lambda - 4\,k}$$ where $k \in \mathbb{R}$ is a constant of integration.

R_{00} is eliminated, and the remaining equations become

$$R_{11}: \quad -\frac{3\,k}{a^2} + \Lambda + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,\mathcal{D}_4'}{4\,a^2\,B^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,B\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,a^2\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{22}: \quad -\frac{3\,k}{a^2} + \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \Lambda + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \quad -\frac{3\,k}{a^2} + \Lambda - \frac{\mathcal{D}_4'}{a^2\,r\,B\,\mathcal{D}_4} + \frac{B'\,\mathcal{D}_4'}{4\,a^2\,B^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,B\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,a^2\,B\,\mathcal{D}_4} = \Lambda$$

$\mathcal{D}_4$ is eliminated in $R_{11}+2R_{22}-R_{44}$, which can then be solved for B:

$$R_{11}+2R_{22}-R_{44}: \quad -\frac{6\,k}{a^2} + \frac{2}{a^2\,r^2} - \frac{2}{a^2\,r^2\,B} + 2\,\Lambda + \frac{2\,B'}{a^2\,r\,B^2} = 2\,\Lambda$$

$$B(r) = \frac{1}{1 - k\,r^2 + \frac{\chi}{r}}$$

where χ is a constant of integration tied to the curled 5th dimension.

R_{11} can be considered to be redundant because R_{11}, R_{22}, and R_{44} have become dependent.

Plugging this back in, the equations become

$$R_{22}: \quad -\frac{k}{a^2} + \Lambda - \frac{\chi}{2\,a^2\,r^3} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,\mathcal{D}_4} + \frac{k\,r\,\mathcal{D}_4'}{2\,a^2\,\mathcal{D}_4} - \frac{\chi\,\mathcal{D}_4'}{2\,a^2\,r^2\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \quad -\frac{3\,k}{a^2} + \Lambda - \frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} + \frac{3\,k\,r\,\mathcal{D}_4'}{2\,a^2\,\mathcal{D}_4} - \frac{3\,\chi\,\mathcal{D}_4'}{4\,a^2\,r^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} - \frac{k\,r^2\,(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{\chi\,(\mathcal{D}_4')^2}{4\,a^2\,r\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} + \frac{k\,r^2\,\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} - \frac{\chi\,\mathcal{D}_4''}{2\,a^2\,r\,\mathcal{D}_4} = \Lambda$$

R_{22} can be solved for $\mathcal{D}_4$:

$$\mathcal{D}_4(r) = -\frac{\zeta\left(-r + k\,r^3 - \chi\right)}{r}$$ where ζ is a constant of integration

$$= 1 - k\,r^2 + \frac{\chi}{r}$$ where ζ has been rolled into p_4

This answers our question about $\mathcal{D}_4$ -- it is indeed sensitive to r.

Also, note that

$$B(r) = \frac{1}{\mathcal{D}_4[r]}.$$

The remaining equation is satisfied:

R_{22}: $\Lambda = \Lambda$

We have found the 5D GRW Equations:

$$\mathcal{R}(t) = \frac{4\,\dot{a}[t]^2}{-4k + \Lambda\, a[t]^2} = c^2$$

where curvature k is a constant satisfying $\Lambda\, a(t)^2 - 4k > 0$,

$$B(r) = \frac{1}{\mathcal{D}_4[r]},$$

$$C(r) = 1,$$

$$\mathcal{D}_4(r) = 1 - k r^2 + \frac{\chi}{r} \quad \text{where } \chi \text{ is a constant subject to } kr^2 - \frac{\chi}{r} < 1,$$

and line element

$$ds^2 = c^2\, dt^2 - \frac{a[t]^2}{1 - k r^2 + \frac{\chi}{r}}\, dr^2 - r^2\, a[t]^2\, d\theta^2 - r^2\, a[t]^2\, \text{Sin}[\theta]^2\, d\phi^2$$
$$- \left(1 - k r^2 + \frac{\chi}{r}\right) a[t]^2\, p_4^2\, d\psi_4^2$$

We will show in Section e that $\mathcal{R}(t) = c^2$ is the Friedmann continuity equation. Thus, as explained in the 4D case, these formulas are our preferred way to present the GRW equations.

Compare the $\mathcal{R}(t)$ coefficients in 5D to those in 4D:

$$\mathcal{R}(t) = \frac{3\,\dot{a}[t]^2}{-3k + \Lambda\, a[t]^2}.$$

This hints at a trend that $\mathcal{R}(t) = \frac{(N-1)\,\dot{a}^2}{(1-N)\,k + \Lambda a^2}$ where N = spacetime dimension.

b: Solve $\mathcal{R}(t) = c^2$ to find $a(t)$

Steps for finding $a(t)$ by solving $\frac{4\,\dot{a}[t]^2}{-4k + \Lambda\, a[t]^2} = c^2$ were provided in Case 4D_a.

The result is

$$a(t) = \begin{cases} 2\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\cosh\left(\frac{1}{2}c\sqrt{\Lambda}\,t\right) & \text{if } k > 0 \;\&\; \Lambda \neq 0 \\ 2\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\sinh\left(\frac{1}{2}c\sqrt{\Lambda}\,t\right) & \text{if } k < 0 \;\&\; \Lambda \neq 0 \\ c\,t\,\sqrt{|k|} & \text{if } k < 0 \;\&\; \Lambda = 0 \\ \alpha\, e^{\pm\frac{1}{2}c\,t\,\sqrt{\Lambda}} & \text{if } k = 0 \;\&\; \Lambda \geq 0 \;\&\; \alpha > 0 \text{ is a free constant} \end{cases}$$

c: Kaluza-Klein (KK) bubble-of-nothing

Recall that the 4D constraint of $kr^2 < 1$ is called Reduced Circumference Polar Coordinates (RCPC). The 4D inequality is always true for $k \leq 0$, but for positive k the domain of r is restricted to $0 < r < \frac{1}{\sqrt{k}}$. As $r \to \frac{1}{\sqrt{k}}$ from below, $B(r) \to \infty$ smoothly.

The 5D analog of RCPC is $kr^2 - \frac{\chi}{r} < 1$. The formula for r such that equality holds is the solution to a cubic equation, $kr^3 - r - \chi = 0$, and is a bit unwieldy, so we will just call it r_0.

- For $\chi \geq 0$ the 5D RCPC domain of r is restricted to $0 \leq r < r_0$, approaching r_0 from below. As $r \to r_0$, then $B \to \infty$ and $\mathbb{D}_4 \to 0$.
- For $\chi < 0$, the domain structure requires separate analyses of the cubic's roots.

The reciprocal relationship $B = \frac{1}{\mathbb{D}_4}$ with $\mathbb{D}_4 \to 0$ at r_0, the 'edge' of the 5D RCPC domain, matches the structure of Witten's Kaluza-Klein (KK) bubble-of-nothing instanton, with space portion of the line element

$$ds^2 = \frac{1}{1 - \frac{\alpha}{r^2}} dr^2 + r^2 d\Omega_3{}^2 + \left(1 - \frac{\alpha}{r^2}\right) d\phi^2$$

A detailed example of how making the geometry smooth at the pinch imposes a relationship between p_4 and free parameters (such as k and χ) is provided in the KK bubble section d of case 6D_1. (6D_1 uses p_5 rather than p_4, and k=0 so the formula that is derived is for χ in terms of p_5.) Reading the 6D_1 KK solution may help the reader prepare to understand the descriptive analysis below.

In Witten's solution for 4D, the circle direction and the radial direction carry reciprocal metric functions, and the circle pinches smoothly to zero size at $r^2 = \alpha$ provided the periodicity of ϕ is tuned correctly. Thus, α is not actually a free parameter there; regularity at the pinch fixes it relative to the periodicity of ϕ.

This solution allows an expanding a(t) sourced by Λ and k as opposed to Witten's static Ricci-flat background. This suggests that this solution is a cosmological generalization of the KK bubble. Standard Kaluza-Klein cosmology assumes $\mathbb{D}_4$ is not a function of r in order to keep the compact circle uniform in size across space. In this solution, I do not assume r-independence, and the field equations then force $\mathbb{D}_4$ to be the specific function of r shown above.

Analogous to the KK bubble, the radial line element for this solution is

$$ds^2 = \frac{1}{\mathbb{D}_4[r]}\, dr^2 + \mathbb{D}_4[r]\; p_4{}^2\; d\psi_4{}^2$$

The curled ψ_4-circle has period of 2π. This places a constraint on the radial line element, an equation that relates χ, k, and p_4. The geometry of this solution is made smooth as $r \to r_0$ by treating χ as a dependent quantity, fixed by regularity. The closed form equation for χ in terms of k and p_4 is:

$$r_0 = \frac{\mp 1 + \sqrt{1 + 3\, k\, p_4^2}}{3\, k\, p_4}$$

$$\chi = r_0 \left(-1 + k\, r_0^2\right)$$

At $k = \frac{1}{p_4{}^2}$, the smoothness condition makes $\chi = 0$.

Smooth solutions exist for every $k \geq \frac{-1}{3\, p_4{}^2}$, each with a corresponding χ.

d: Free-gauge GRW

The 5D FG-GRW equations are shown in the table below.

		4D	5D
GRW	Constraints	None	None
	KK-bubble Con-straints (optional)	n/a	$\chi = r_0\left(-1 + k r_0{}^2\right)$ where $r_0 = \frac{\mp 1 + \sqrt{1 + 3\, k\, p_4^2}}{3\, k\, p_4}$ and $k \geq -\frac{1}{3\, p_4{}^2}$
	$\mathcal{A}$	$\mathcal{A}(t) = c^2$	$\mathcal{A}(t) = c^2$
	B	$\frac{1}{1 - k\, r^2}$	$\frac{1}{1 - k\, r^2 + \frac{\chi}{r}}$
	C	1	1
	$\mathbb{D}_4$		$1 - k\, r^2 + \frac{\chi}{r} = \frac{1}{B}$
FG GRW	$\mathcal{A}$	$\frac{3\, \dot{a}^2}{-3\, k + a^2\, \Lambda}$	$\frac{4\, \dot{a}^2}{-4\, k + a^2\, \Lambda}$
	B	$\frac{(2\, C + r\, C')^2}{4\, C\, (1 - k\, r^2\, C)}$	$\frac{(2\, C + r\, C')^2}{4\, C\, \left(1 - k r^2\, C + \frac{\chi}{r\, \sqrt{C}}\right)} = \frac{(2\, C + r\, C')^2}{4\, C \mathbb{D}_4}$
	C	C (free)	C (free)
	$\mathbb{D}_4$		$1 - k\, r^2\, C + \frac{\chi}{r\, \sqrt{C}}$

Notes:

- $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation
  - ✓ The solution to the equation is the a(t) formula from Section b
- The two constraint rows display restrictions on parameter constants
  - ✓ Due to the field equations
  - ✓ Due to KK-bubble smoothness
- By contrast, domain restrictions on r (RCPC and its analogs) are not displayed as constraints
- FG-GRW has the same constraints (both rows) as GRW

The FG-GRW results were obtained by using the pattern in 4D between FLRW and FG-FLRW:

$\mathcal{A}$: Use the $\mathcal{A}(t)$ formula found in GRW

B: Replace $r^2 \to r^2 C$ and multiply result by $\frac{(2C + rC')^2}{4C}$

C: Replace C=1 by free parameter C

$\mathcal{D}_4$ is new, so there is no established pattern. What works is simply

$\mathcal{D}_4$: Replace r^2 by $r^2 C$ (and r by $r\sqrt{C}$)

The line element is

$$ds^2 = \frac{4\,\dot{a}[t]^2}{-4k + \Lambda\, a[t]^2}\,dt^2 - \frac{a[t]^2\,(2C[r] + rC'[r])^2}{4C[r]\left(1 + \frac{\chi}{r\sqrt{C[r]}} - k r^2 C[r]\right)}\,dr^2 - r^2 a[t]^2 C[r]\,d\Theta^2$$
$$- r^2 a[t]^2 \mathrm{Sin}[\Theta]^2 C[r]\,d\phi^2$$
$$- a[t]^2 p_4^2 \left(1 + \frac{\chi}{r\sqrt{C[r]}} - k r^2 C[r]\right) d\psi_4^2$$

To prove that this is indeed an exact solution, we
merely have to confirm that the field equations are satisfied:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: $0 = 0$

R_{03}: $0 = 0$

R_{04}: $0 = 0$

R_{11}: $\Lambda = \Lambda$

R_{12}: $0 = 0$

R_{13}: $0 = 0$

R_{14}: $0 = 0$

R_{22}: $\Lambda = \Lambda$

R_{23}: $0 = 0$

R_{24}: $0 = 0$

R_{33}: $\Lambda = \Lambda$

R_{34}: $0 = 0$

R_{44}: $\Lambda = \Lambda$

e: The Vacuum Friedmann Equations

Plugging $\mathcal{A}$, B, C, and D_4 (but not a) from the GRW solution into the field equations yields:

R_{00}: $\frac{4\,\ddot{a}}{a\,c^2} = \Lambda$

R_{11}: $\frac{3\,k}{a^2} + \frac{3\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}: $\frac{3\,k}{a^2} + \frac{3\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$-\frac{1}{12}R_{00}+\frac{1}{3}R_{11}$: $\frac{k}{a^2} + \frac{\dot{a}^2}{a^2\,c^2} = \frac{\Lambda}{4}$

$\frac{1}{5}R_{00}+\frac{1}{5}R_{11}$: $\frac{3\,k}{5\,a^2} + \frac{3\,\dot{a}^2}{5\,a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \frac{2\,\Lambda}{5}$

As pointed out earlier, $\mathcal{A}(t) = c^2$ is simply a rearrangement of the continuity equation. Since the $a(t)$ formula in Part b is the result of solving $\mathcal{A}(t) = c^2$, the 4-branch $a(t)$ formula is also the solution to the Friedmann equations.